EASY · ADVANCED MATH

Nonlinear functions

06

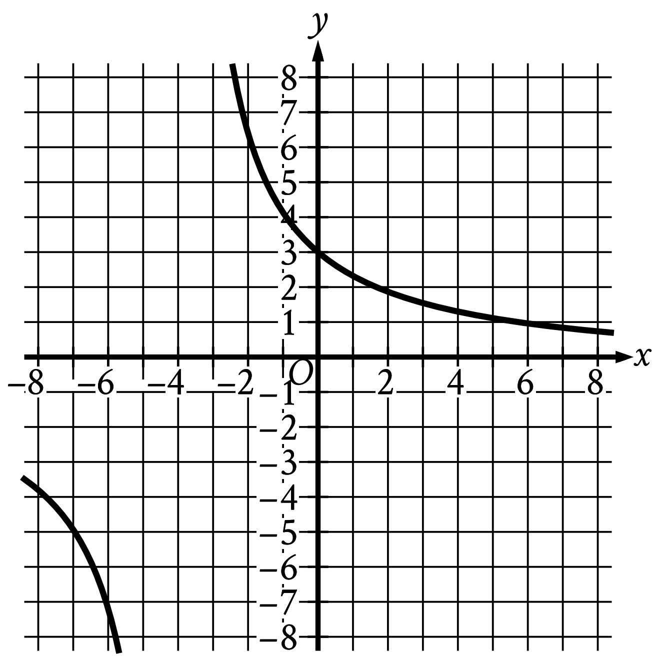
10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1

GRID-IN ADVANCED MATH



- For the first curve:
  - Moving from left to right:
    - The curve is in quadrant 3.
    - As x decreases, the curve approaches the line y equals 0.
    - As x increases, the curve approaches the line x equals negative 4.
- For the second curve:
  - Moving from left to right:
    - The curve passes from quadrant 2 to quadrant 1.
    - As x decreases, the curve approaches the line x equals negative 4.
    - As x increases, the curve approaches the line y equals 0.
- The 2 curves pass through the following points:
  - approximately (negative 7 comma negative 4.9)
  - (0 comma 3)
  - approximately (7 comma 0.8)

The graph of $y = f x$ is shown in the xy-plane. The value of $f 0$ is an integer. What is the value of $f 0$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

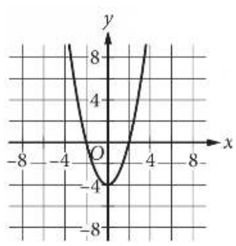
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

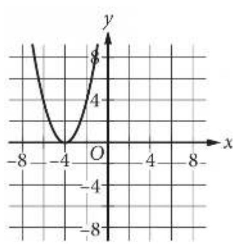
Q2

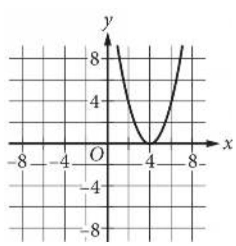
ADVANCED MATH

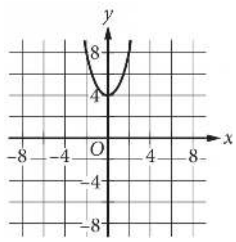
$$f(x) = x^2 + 4$$

The function f is defined as shown. Which of the following graphs in the xy-plane could be the graph of $y = f(x)$?

A. 

B. 

C. 

D. 

Q3

ADVANCED MATH

The function $f\left(x\right) = 240{,}0001.22^{x}$ gives a company's predicted annual revenue, in dollars, x years after the company started selling jewelry online, where $0 < x \leq 10$. What is the best interpretation of the statement "$f5$ is approximately equal to $648{,}650$" in this context?

A. 5 years after the company started selling jewelry online, its predicted annual revenue is approximately $648{,}650$ dollars.

B. 5 years after the company started selling jewelry online, its predicted annual revenue will have increased by a total of approximately $648{,}650$ dollars.

C. When the company's predicted annual revenue is approximately $648{,}650$ dollars, it is 5 times the predicted annual revenue for the previous year.

D. When the company's predicted annual revenue is approximately $648{,}650$ dollars, it is 5% greater than the predicted annual revenue for the previous year.

Q4

ADVANCED MATH

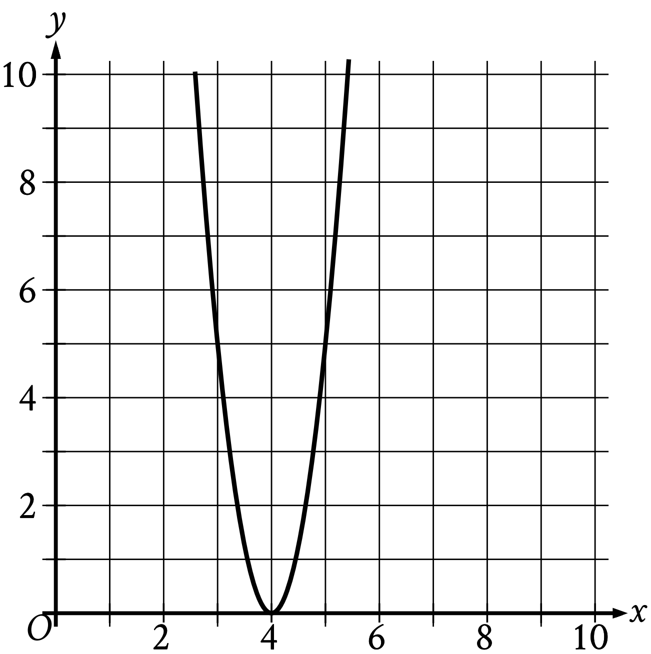
The function f is defined by $f x = 5x^2$. What is the value of $f 8$?

A. 40

B. 50

C. 80

D. 320



- The parabola opens upward.
- The vertex is at point (4 comma 0).
- The parabola passes through the following points:
    - (3 comma 5)
    - (4 comma 0)
    - (5 comma 5)

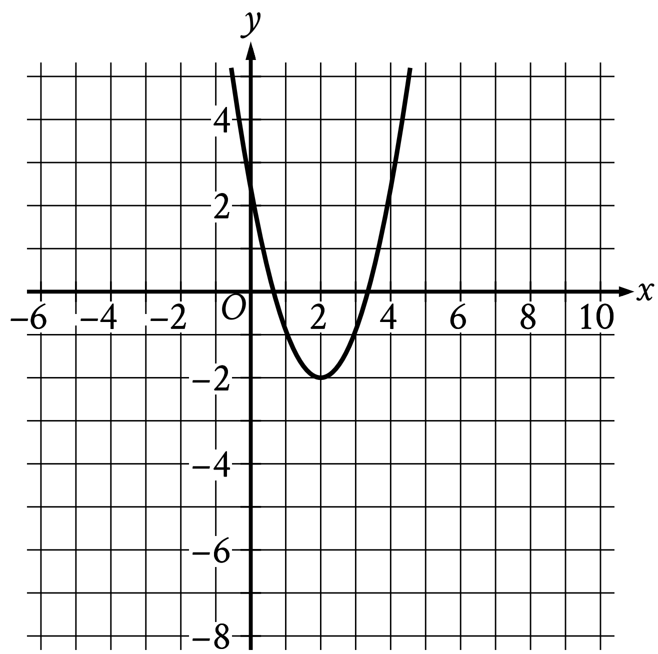
What is the x-intercept of the graph shown?

A. -5, 0

B. 5, 0

C. -4, 0

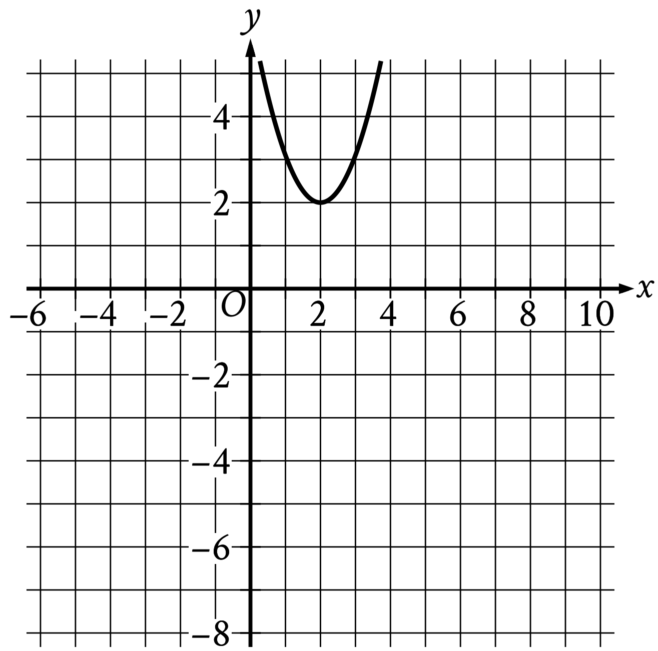
D. 4, 0



- The parabola opens upward.
- The vertex is at the point (2 comma negative 2).
- The parabola passes through the following points:
  - (1 comma negative nine tenths)
  - (2 comma negative 2)
  - (3 comma negative nine tenths)

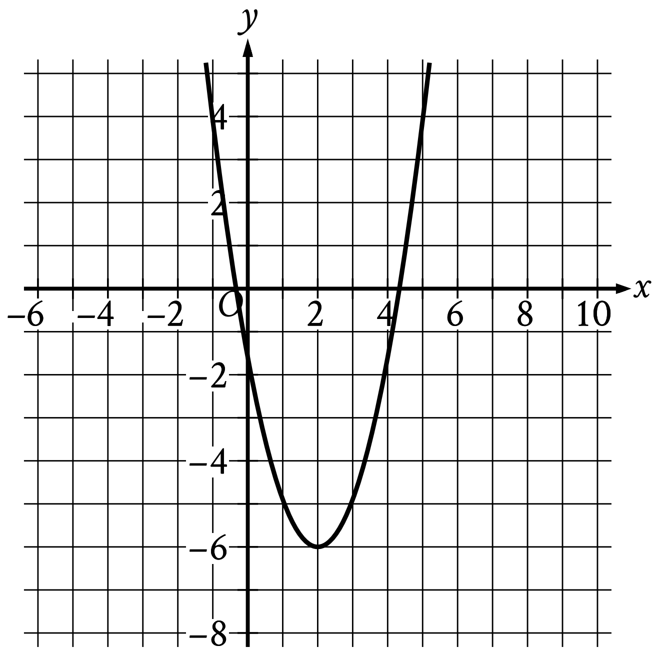
The graph shown will be translated up 4 units. Which of the following will be the resulting graph?

A.



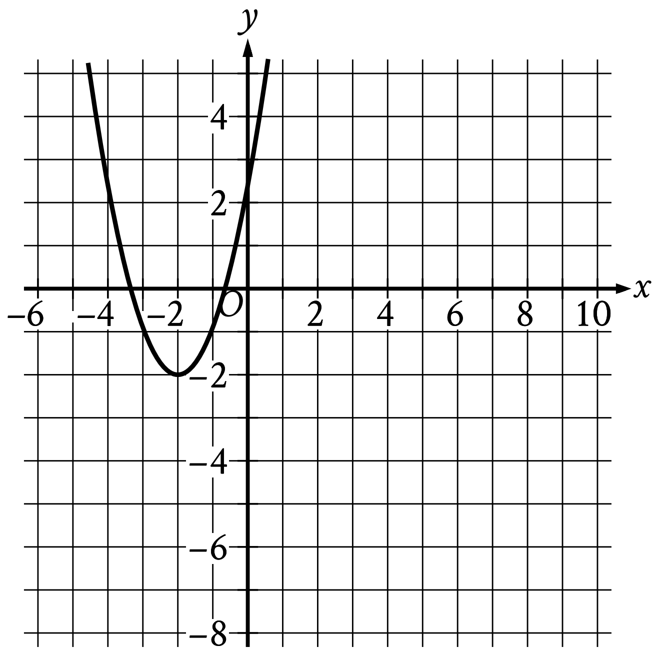
- The parabola opens upward.
- The vertex is at the point (2 comma 2).
- The parabola passes through the following points:
  - (1 comma StartFraction 31 Over 10 EndFraction)
  - (2 comma 2)
  - (3 comma StartFraction 31 Over 10 EndFraction)

B.



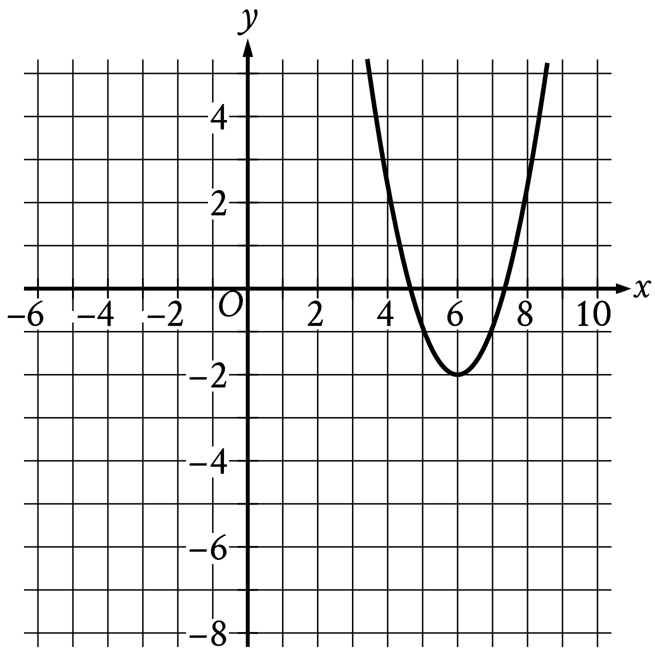
- The parabola opens upward.
- The vertex is at the point (2 comma negative 6).
- The parabola passes through the following points:
  - (1 comma negative StartFraction 49 Over 10 EndFraction)
  - (2 comma negative 6)
  - (3 comma negative StartFraction 49 Over 10 EndFraction)

C.



- The parabola opens upward.
- The vertex is at the point (negative 2 comma negative 2).
- The parabola passes through the following points:
  - (negative 3 comma negative nine tenths)
  - (negative 2 comma negative 2)
  - (negative 1 comma negative nine tenths)

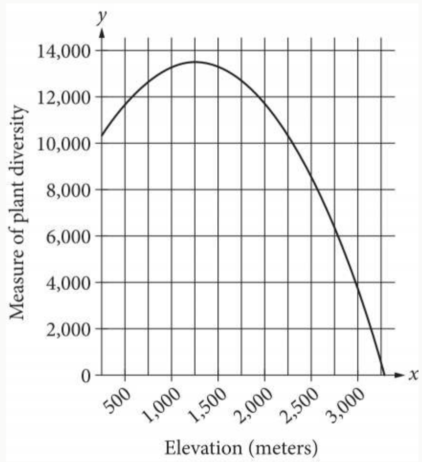
D.



- The parabola opens upward.
- The vertex is at the point (6 comma negative 2).
- The parabola passes through the following points:
    - (5 comma negative nine tenths)
    - (6 comma negative 2)
    - (7 comma negative nine tenths)

Q7

ADVANCED MATH



The quadratic function graphed above models a particular measure of plant diversity as a function of the elevation in a region of Switzerland. According to the model, which of the following is closest to the elevation, in meters, at which plant diversity is greatest?

A. 13,500

B. 3,000

C. 1,250

D. 250

Q8

ADVANCED MATH

$$f(x) = (x + 0.25x)(50 - x)$$

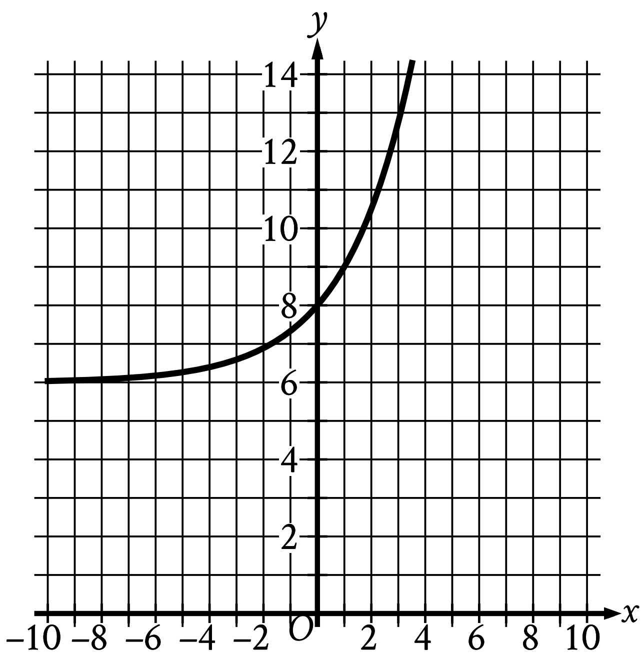
The function f is defined above. What is the value of $f(20)$?

A. 250

B. 500

C. 750

D. 2,000



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up gradually to point (0 comma 8).
  - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
  - (0 comma 8)
  - (1 comma 9)

What is the y-intercept of the graph shown?

A. -8, 0

B. -6, 0

C. 0, 6

D. $0, 8$

Q10 ADVANCED MATH

There are 240 players in a tennis competition that includes 4 rounds of matches. Each player in the competition will play a match against another player in round 1. At the end of each round, the player who loses the match is eliminated and the player who won the match advances to the next round to play a match against another winning player. Which equation gives the number of players, p, eliminated at the end of round r, where $r \leq 4$?

A. $p = 15\frac{1}{2}^{r}$

B. $p = 152^{r}$

C. $p = 240\frac{1}{2}^{r}$

D. $p = 2402^{r}$